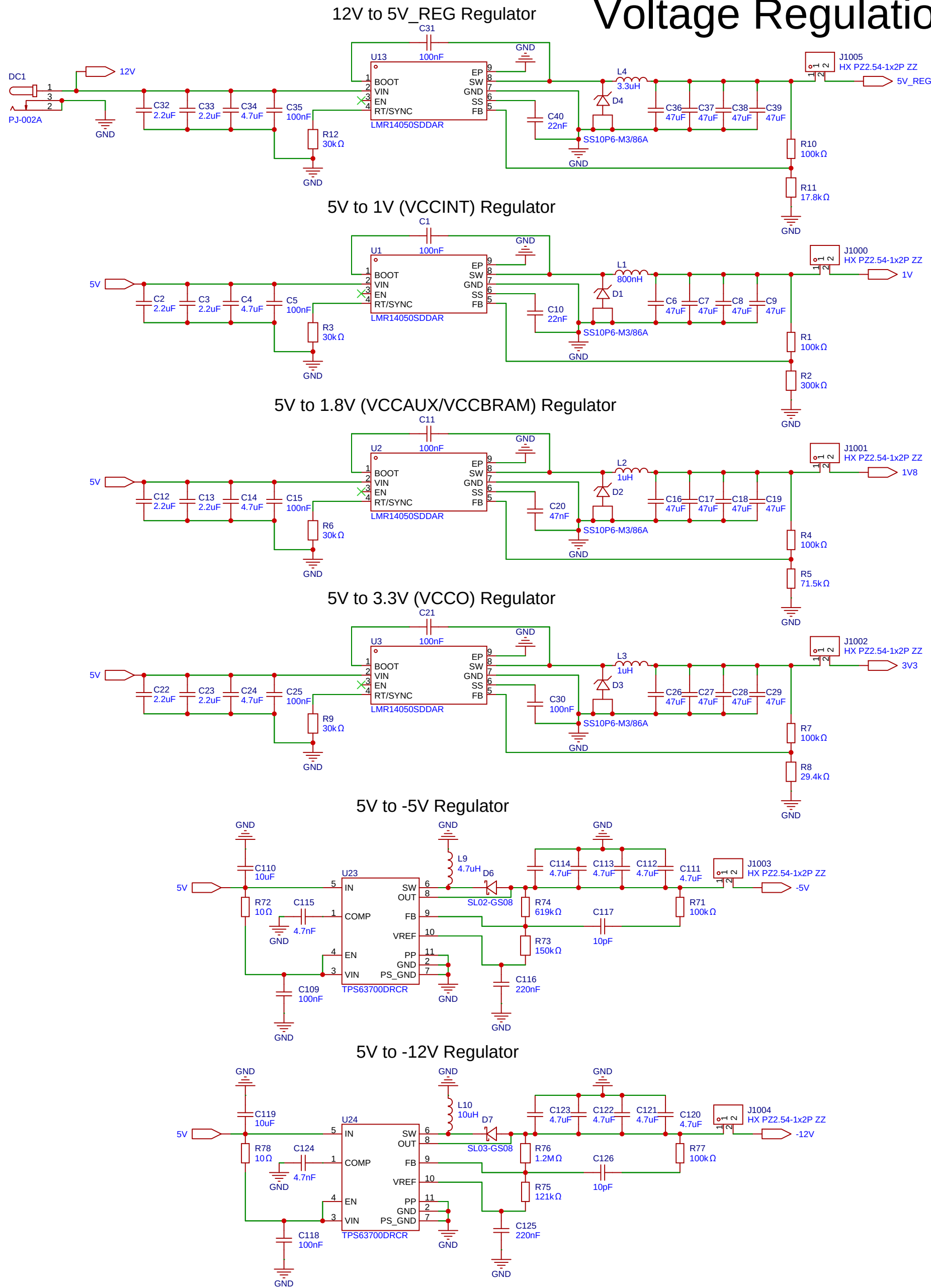
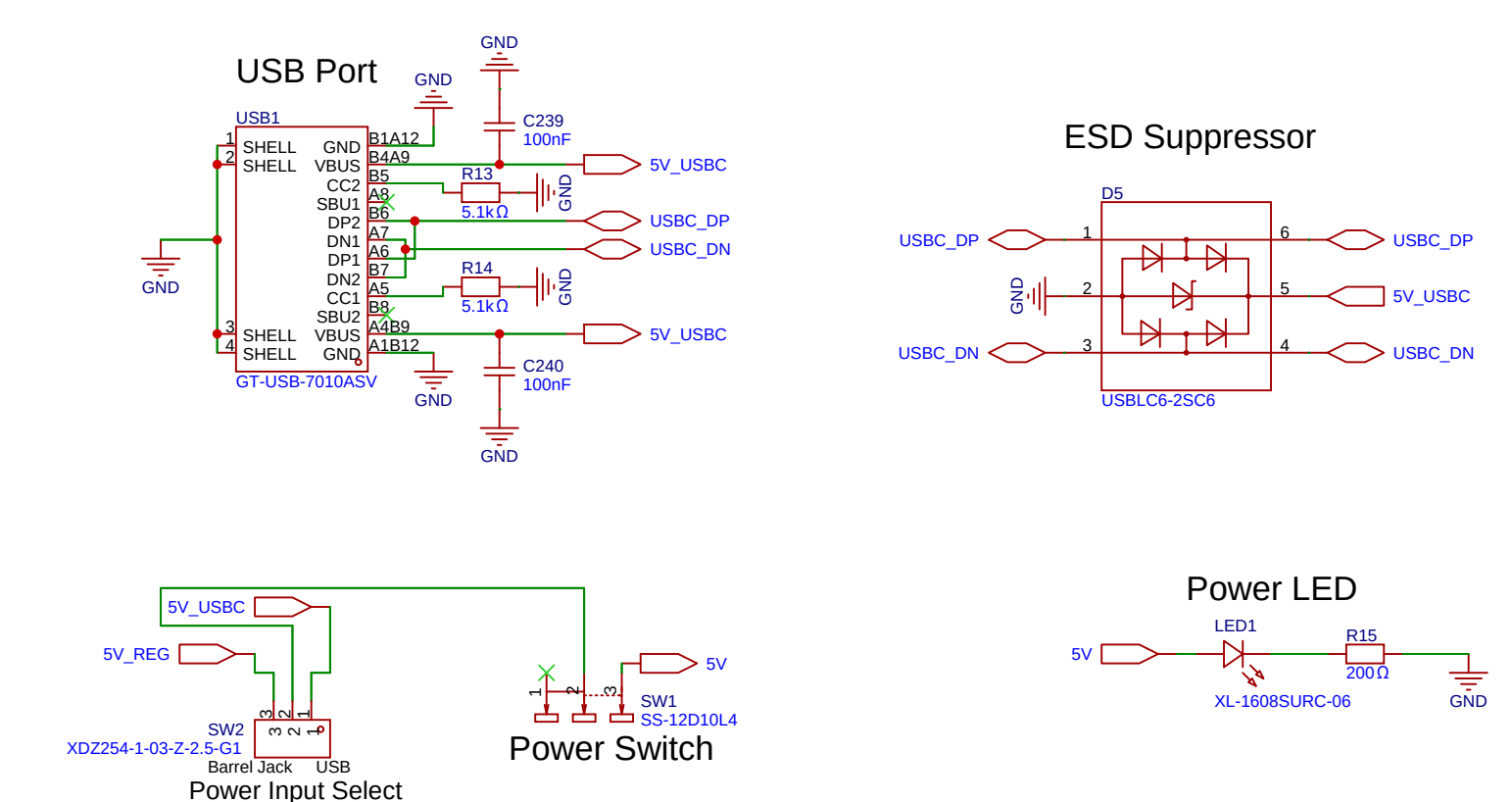


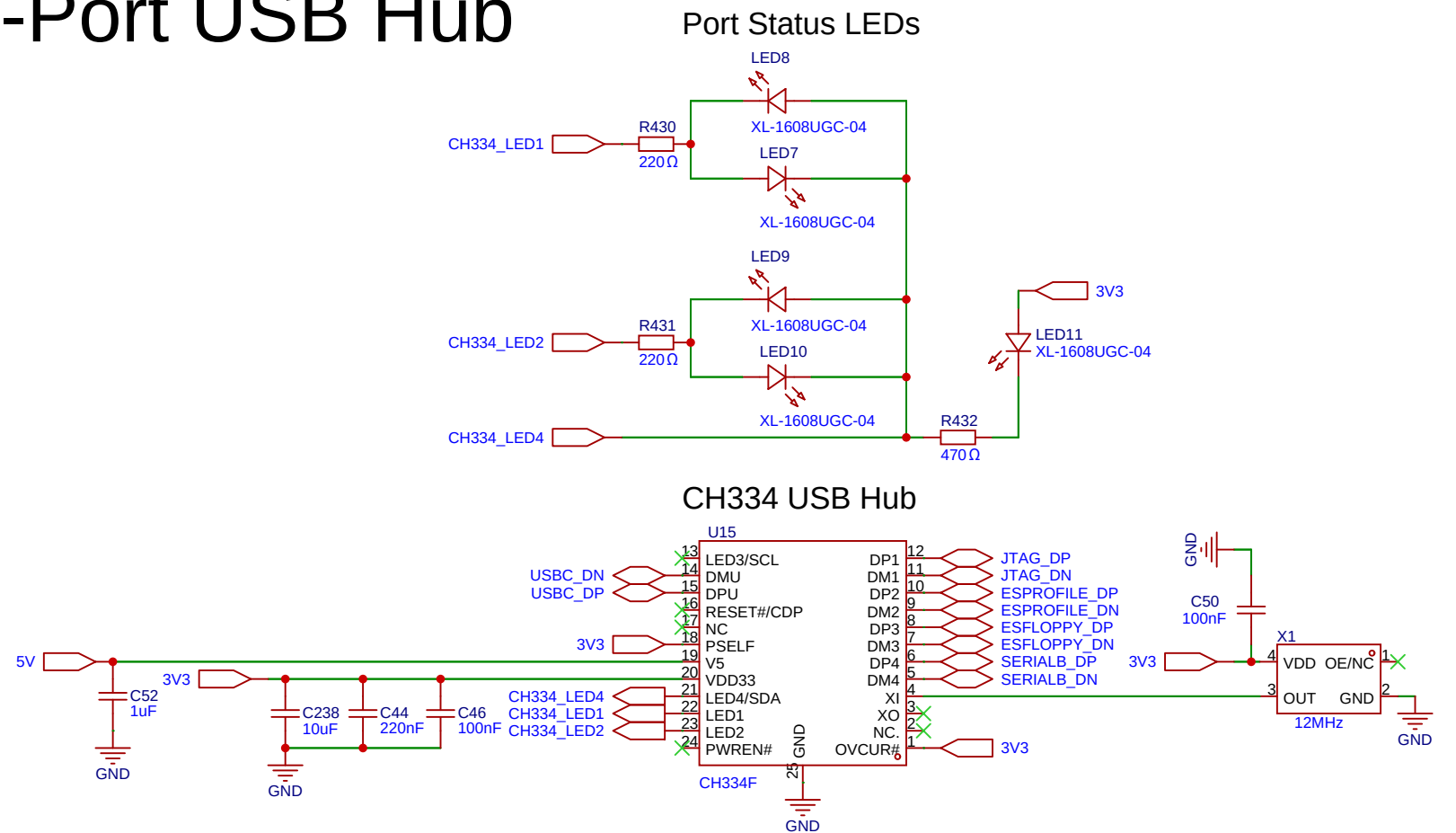
Voltage Regulation



USB Port and Power Switching

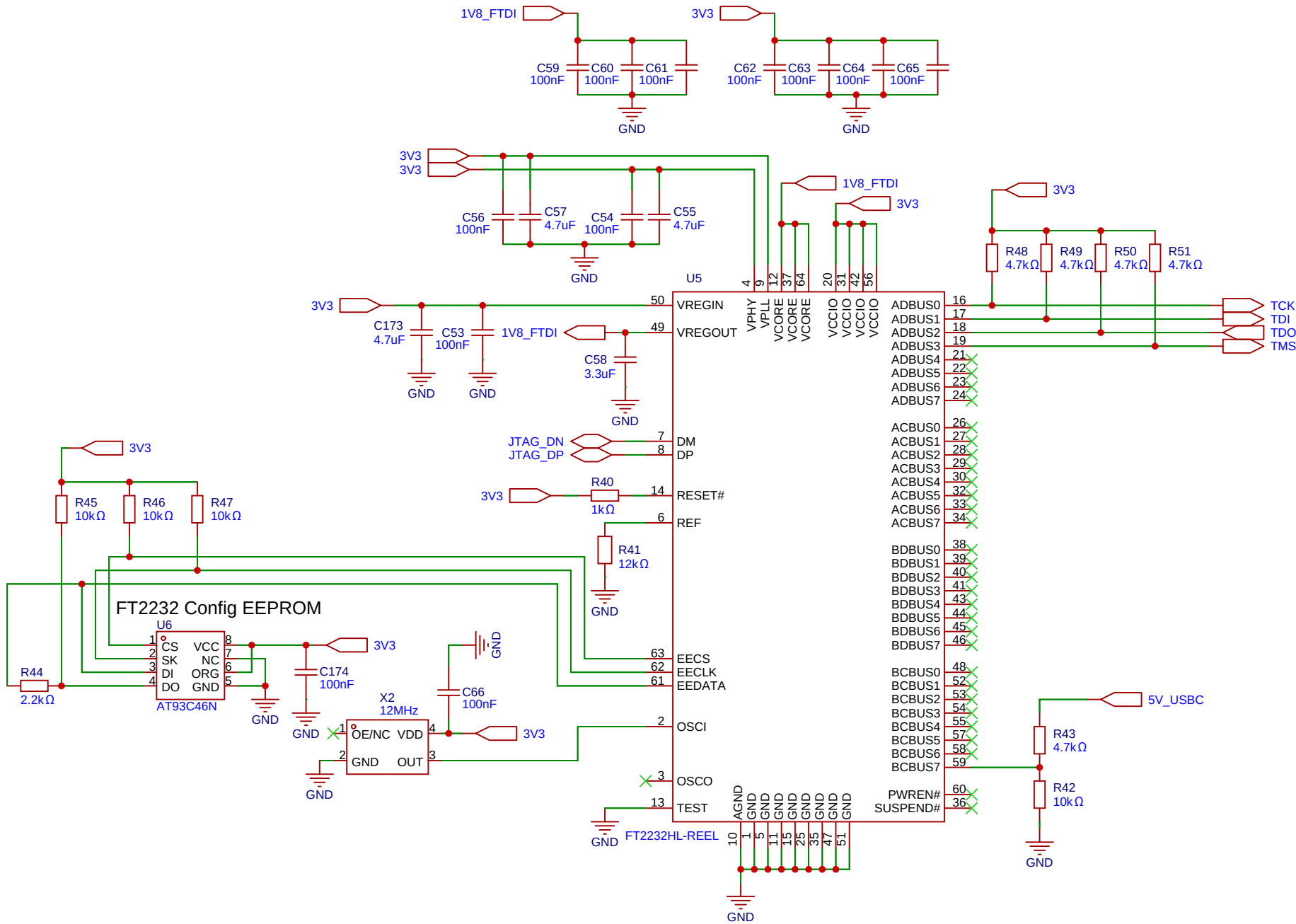


4-Port USB Hub

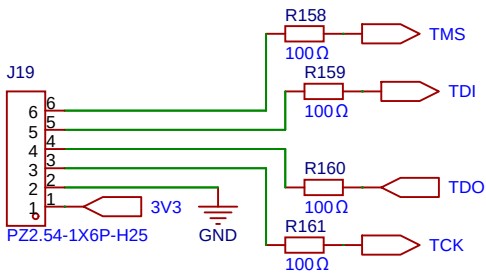


Schematic	LisaFPGA Schematic		Create at	2025-10-14
Board	LisaFPGA		Update at	2025-11-01
Drawn	LisaFPGA		Page	Power and USB Input
Reviewed				
		Version	Size	Page 1 Total 8
		V1.0	A4	EasyEDA.com

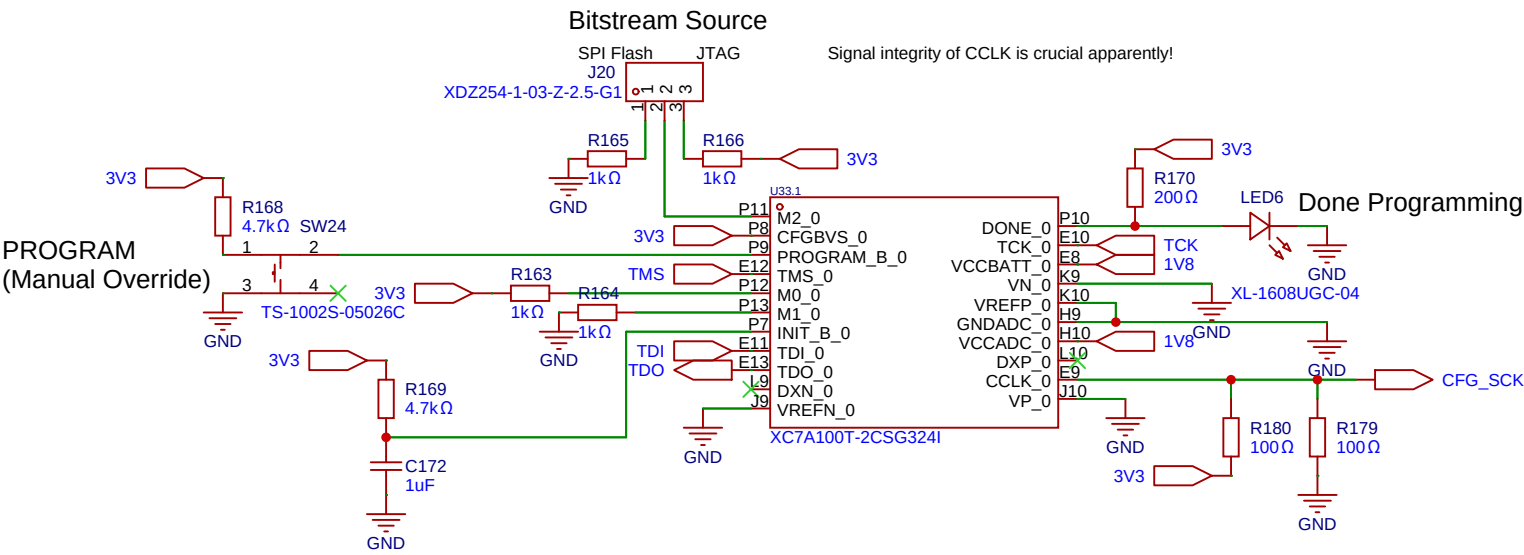
FT2232 USB to JTAG



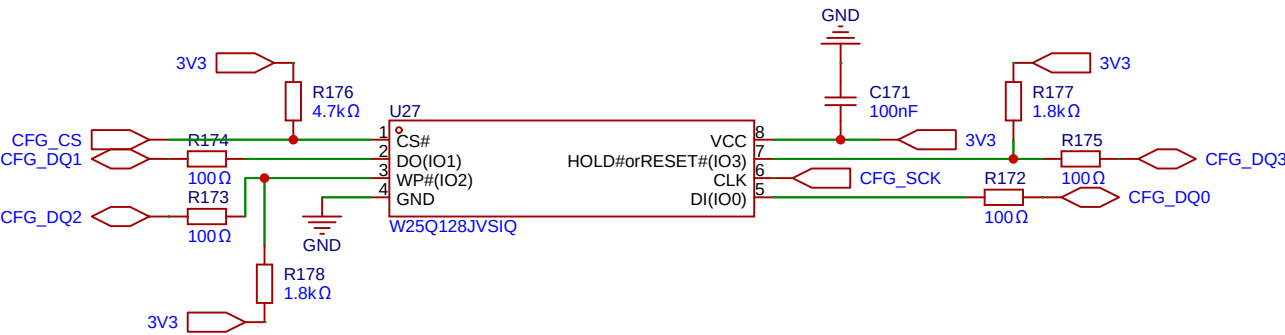
Dedicated JTAG Header



FPGA Programming/Config Interface

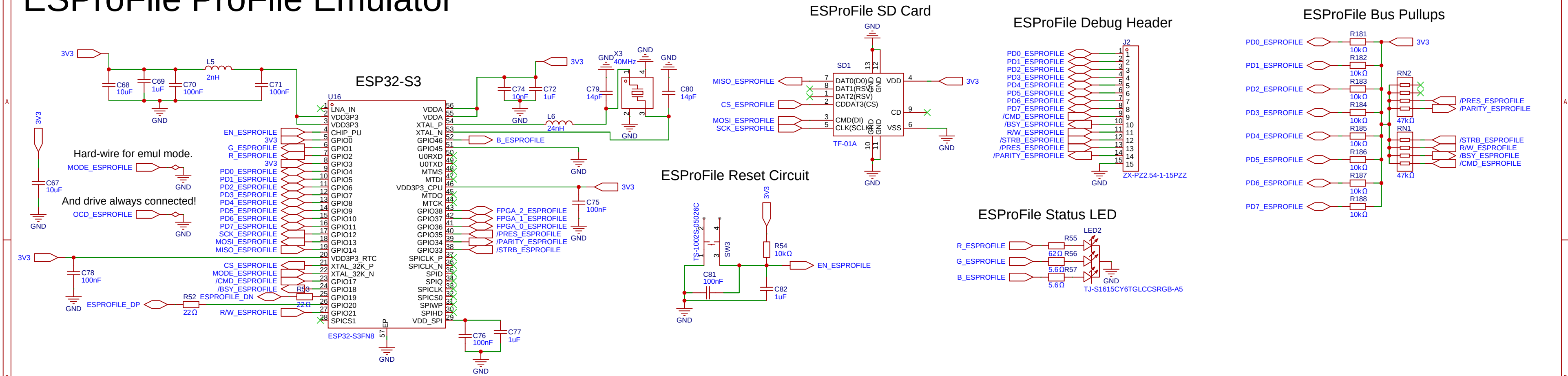


Bitstream SPI Flash

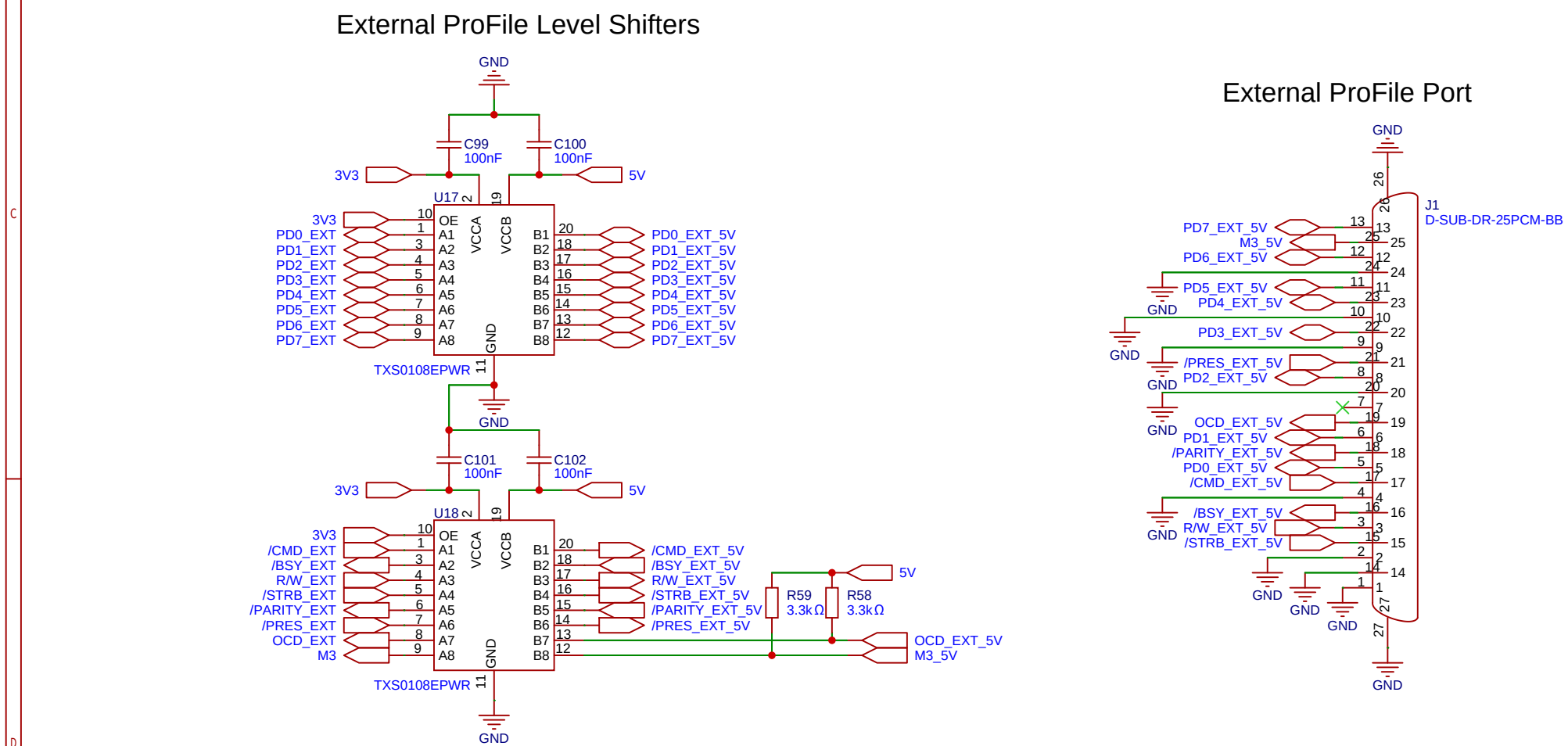


Schematic	LisaFPGA Schematic		Create at	2025-10-15
Board	LisaFPGA		Update at	2025-10-30
Drawn		LisaFPGA		
Reviewed				
		Version	Size	Page 2 Total 8
EasyEDA		V1.0	A4	EasyEDA.com

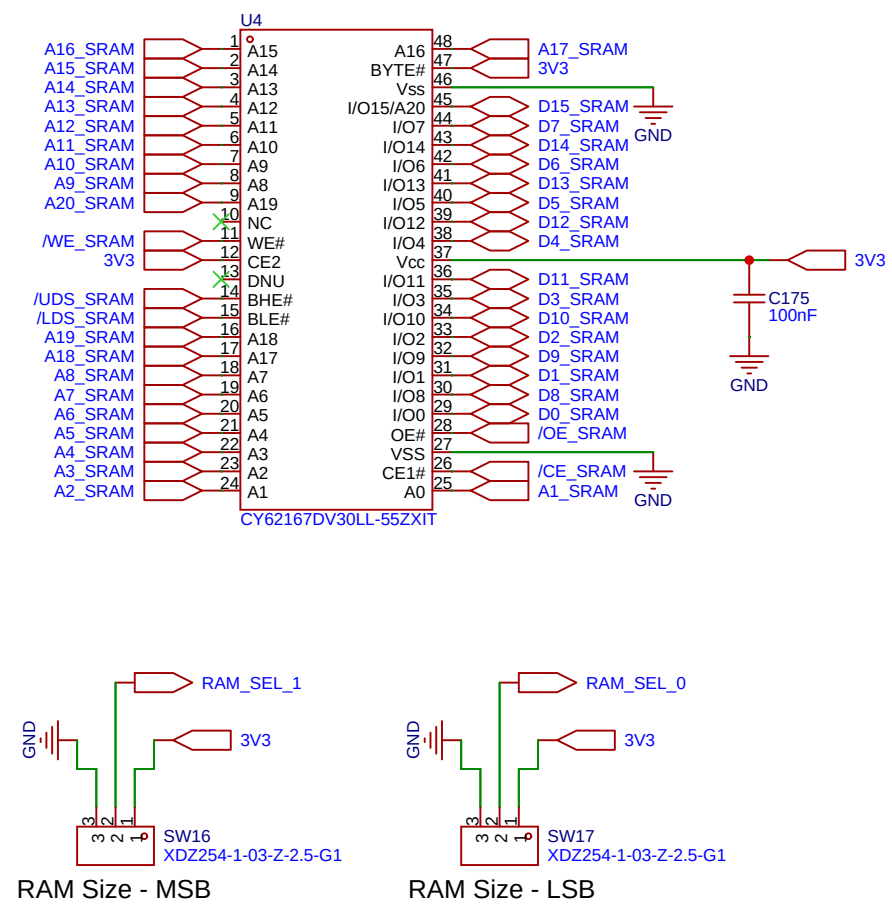
ESProFile ProFile Emulator



External ProFile Connection

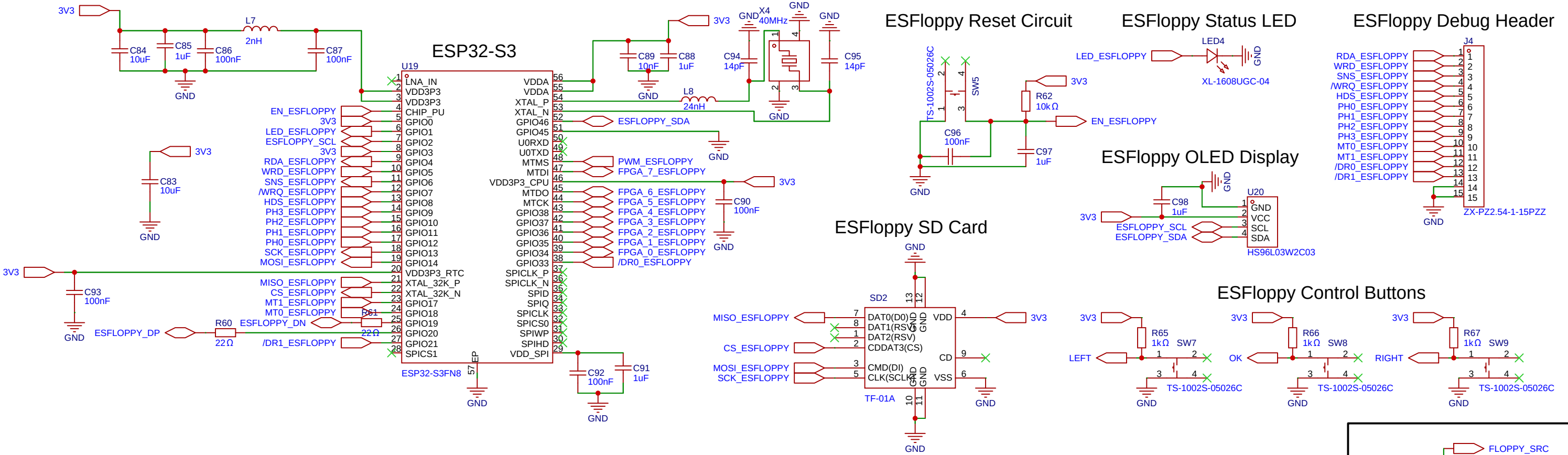


Lisa System Memory

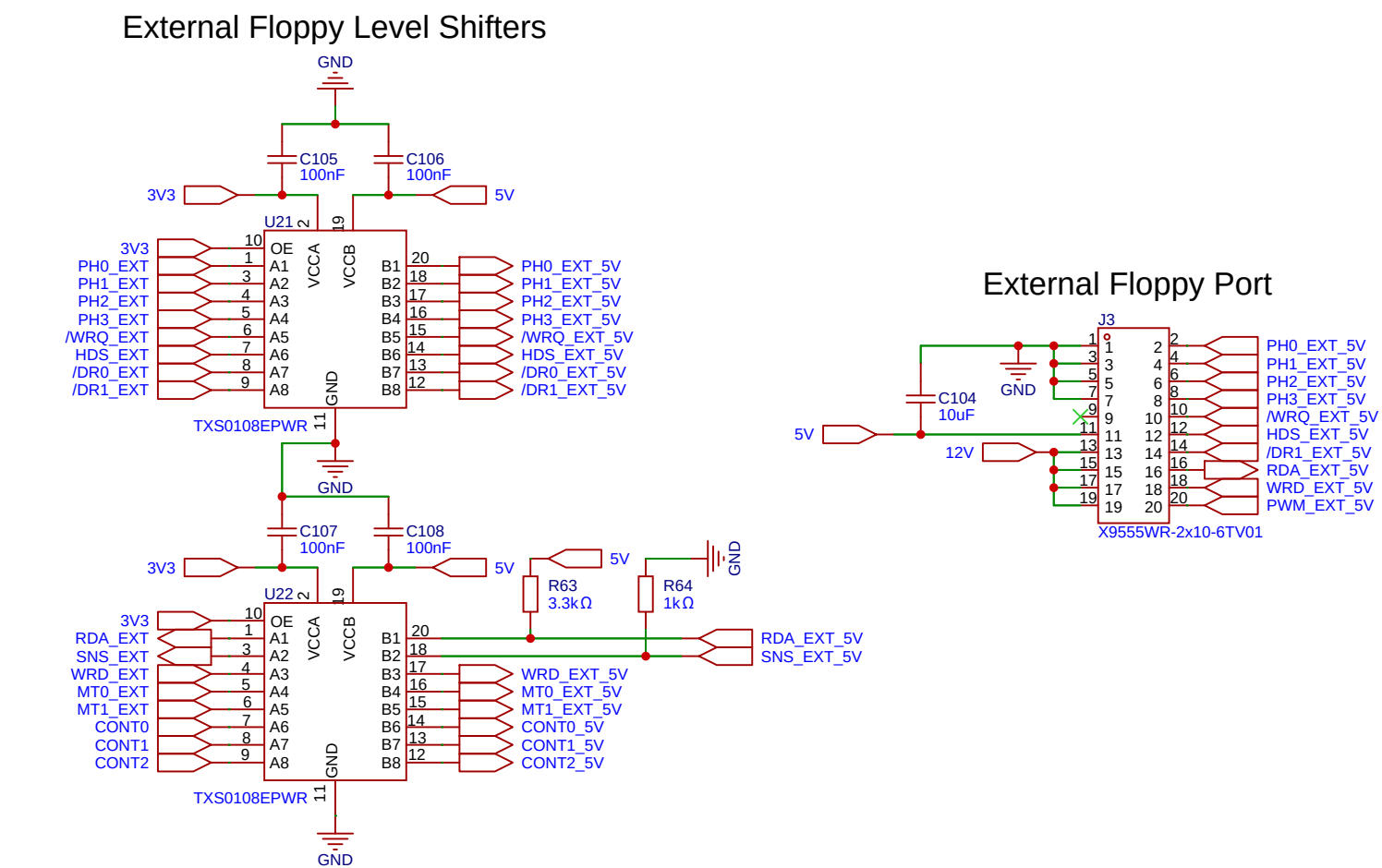


Schematic	LisaFPGA Schematic			Create at	2025-10-24
				Update at	2025-10-31
Board	LisaFPGA			Page	ProFile Interface and SRAM
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 3 Total 8	
		V1.0	A4	EasyEDA.com	

ESFloppy Floppy Emulator



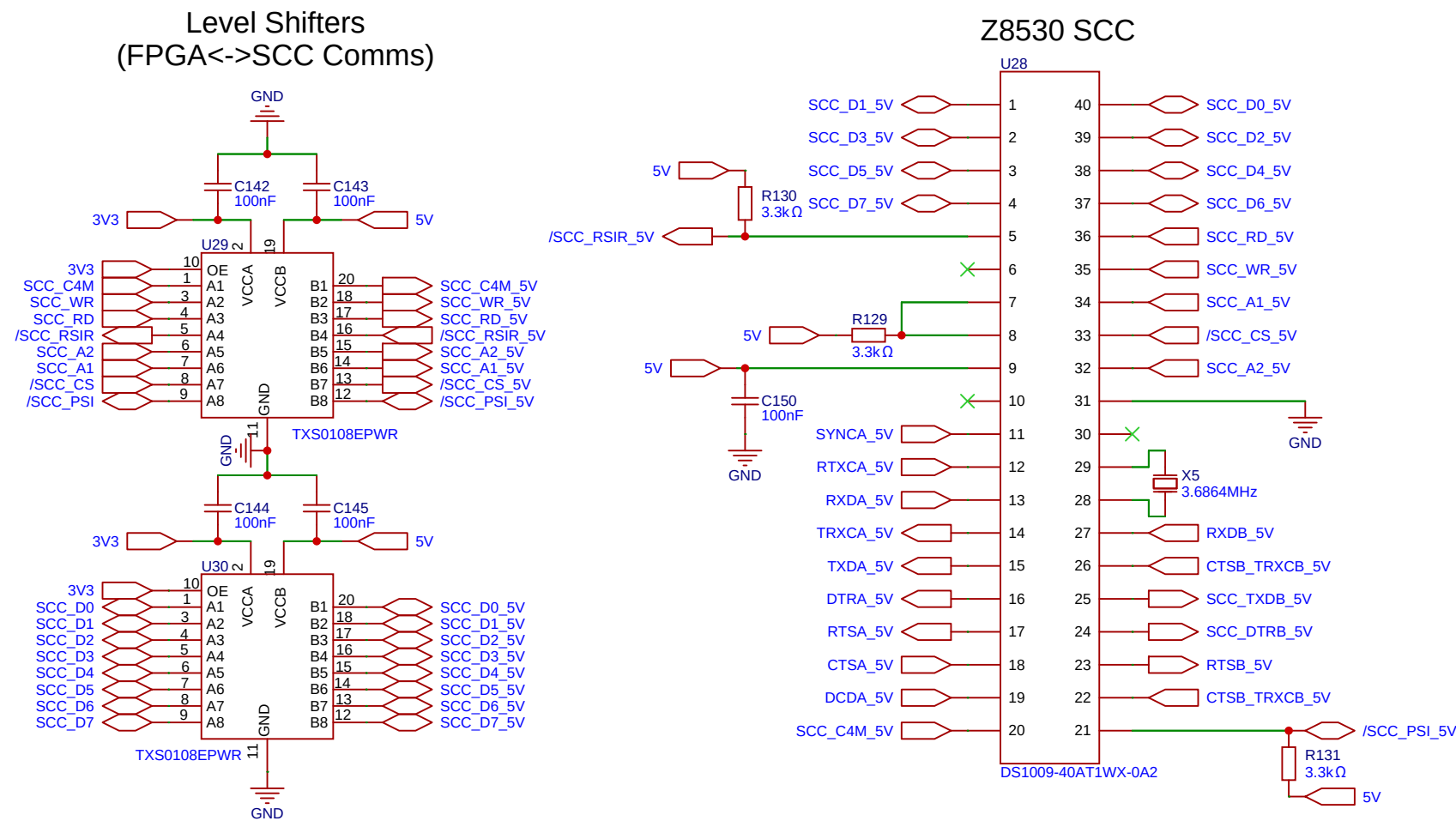
External Floppy Connection



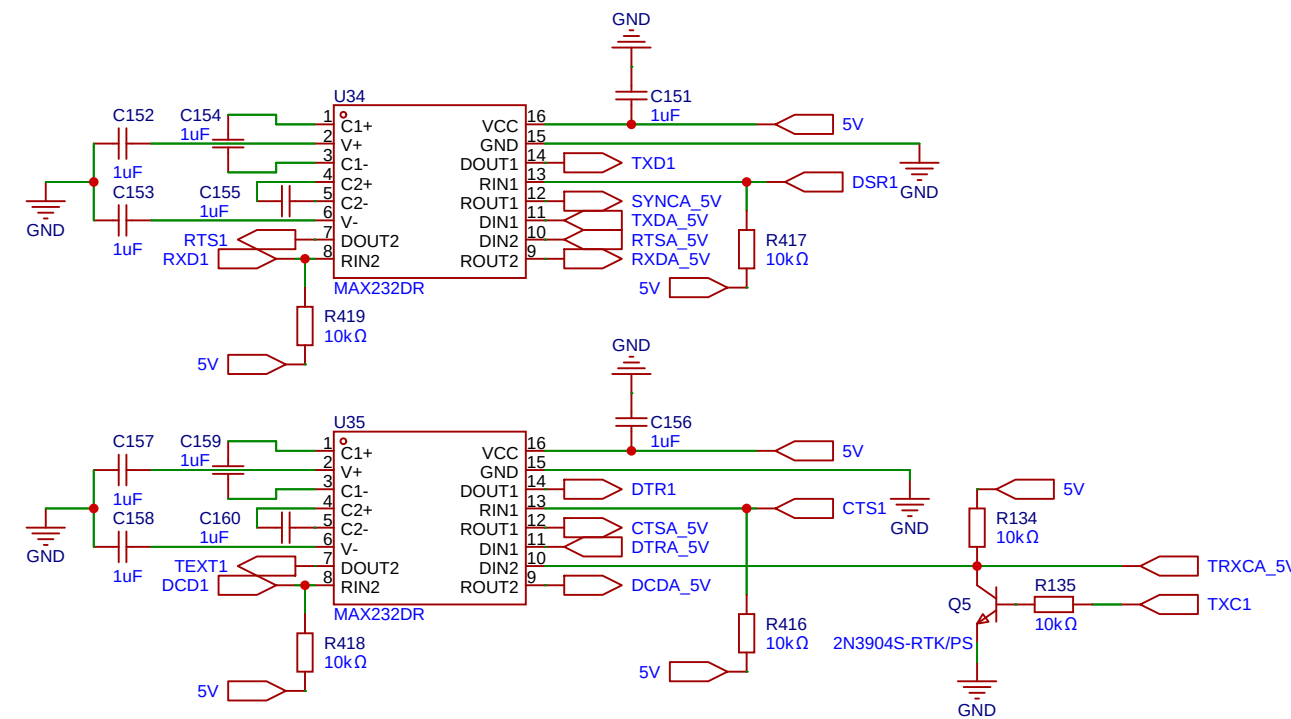
Well, that was some poor planning.
There's nothing to fill this space.
Whoops.

Schematic	LisaFPGA Schematic			Create at	2025-10-27
				Update at	2025-10-31
Board	LisaFPGA			Page	Floppy Interface
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 4 Total 8	
		V1.0	A4	EasyEDA.com	

The SCC Itself!

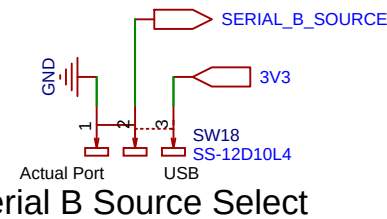
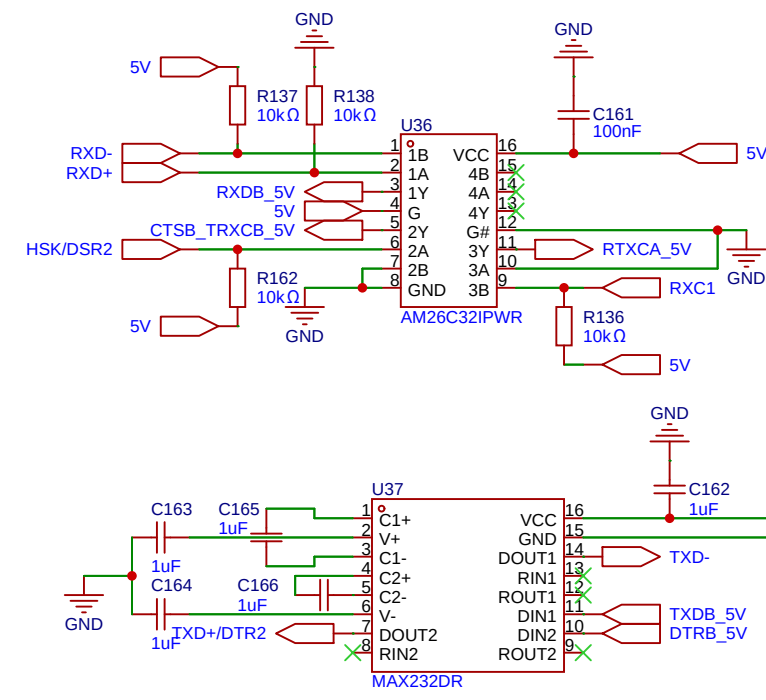


Serial A Transceivers

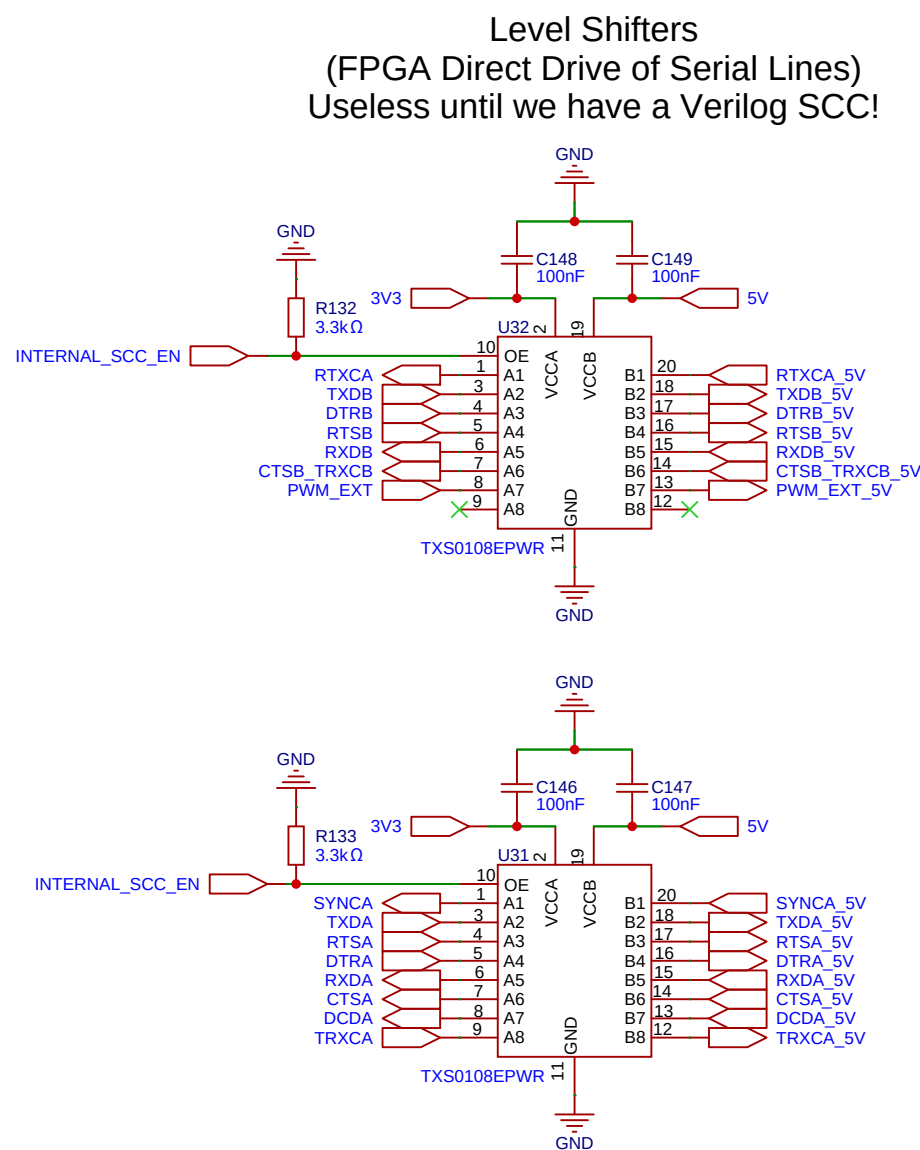


Serial B Transceivers

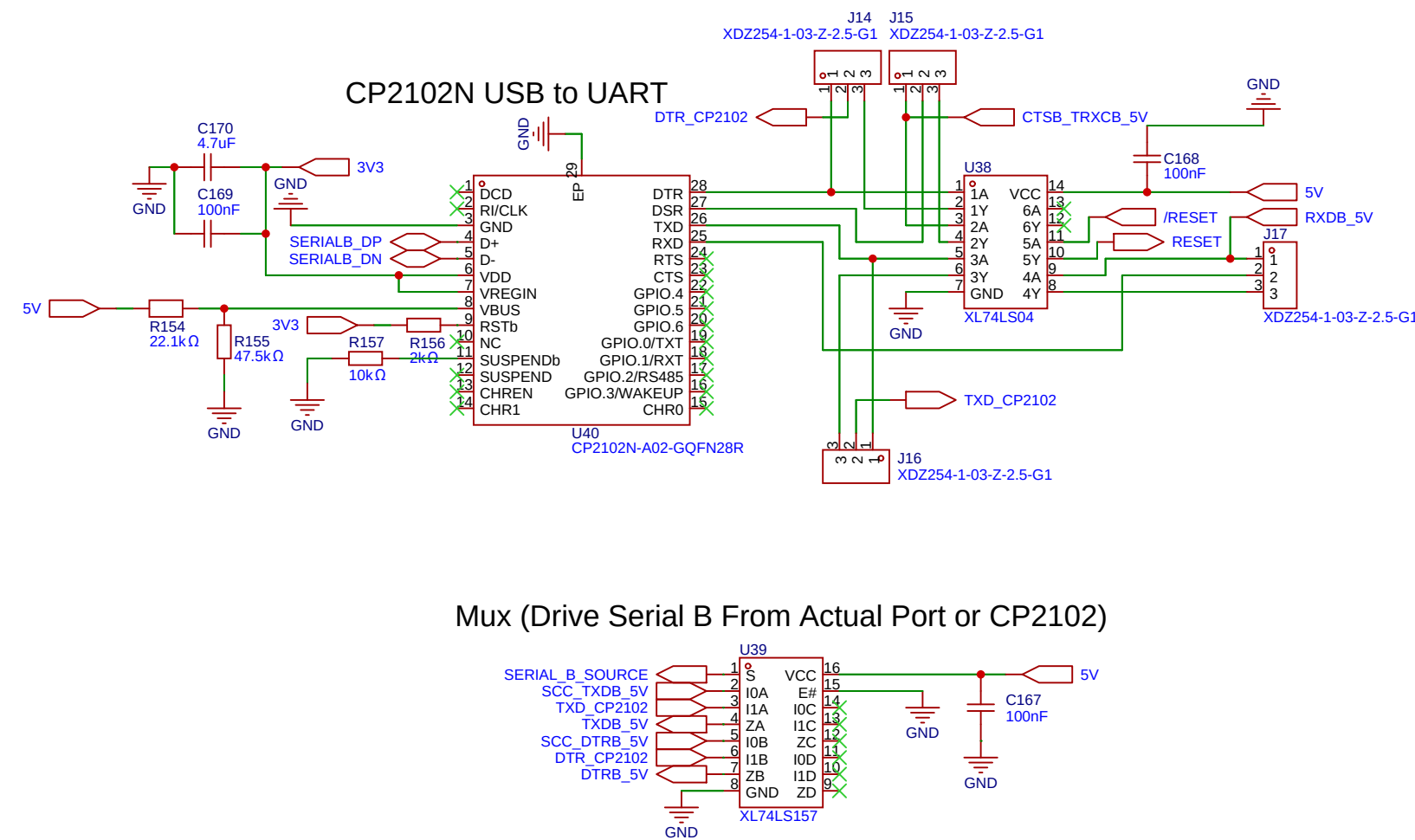
A note about Serial B: The TXD LINES DONT SUPPORT DIFFERENTIAL OPERATION!
Serial B can operate in 2 modes: differential (RS-422) and single-ended (RS-232).
But we can't do differential because I couldn't find a modern line driver IC that could be switched between differential and single-ended on the fly like the Lisa requires.
So I instead used a single-ended IC drivers since you'll be using the port in RS-232 mode 99 times out of 100 anyway.
The RXD lines do still support differential mode though, so at least you can receive differential data!



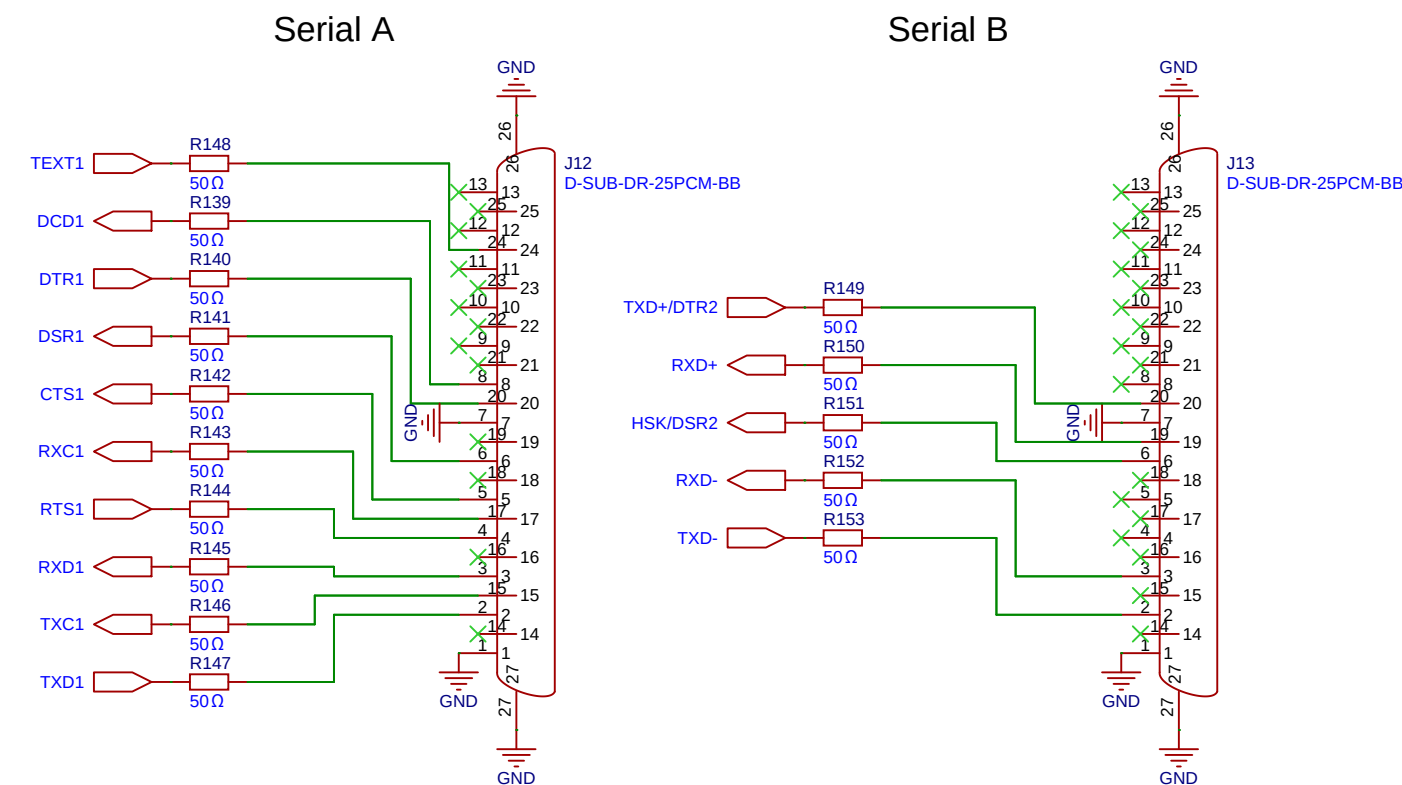
Serial I/O to FPGA



USB to Serial B

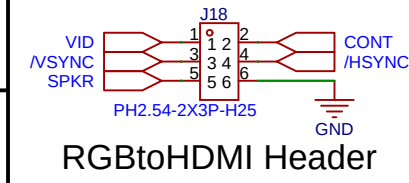
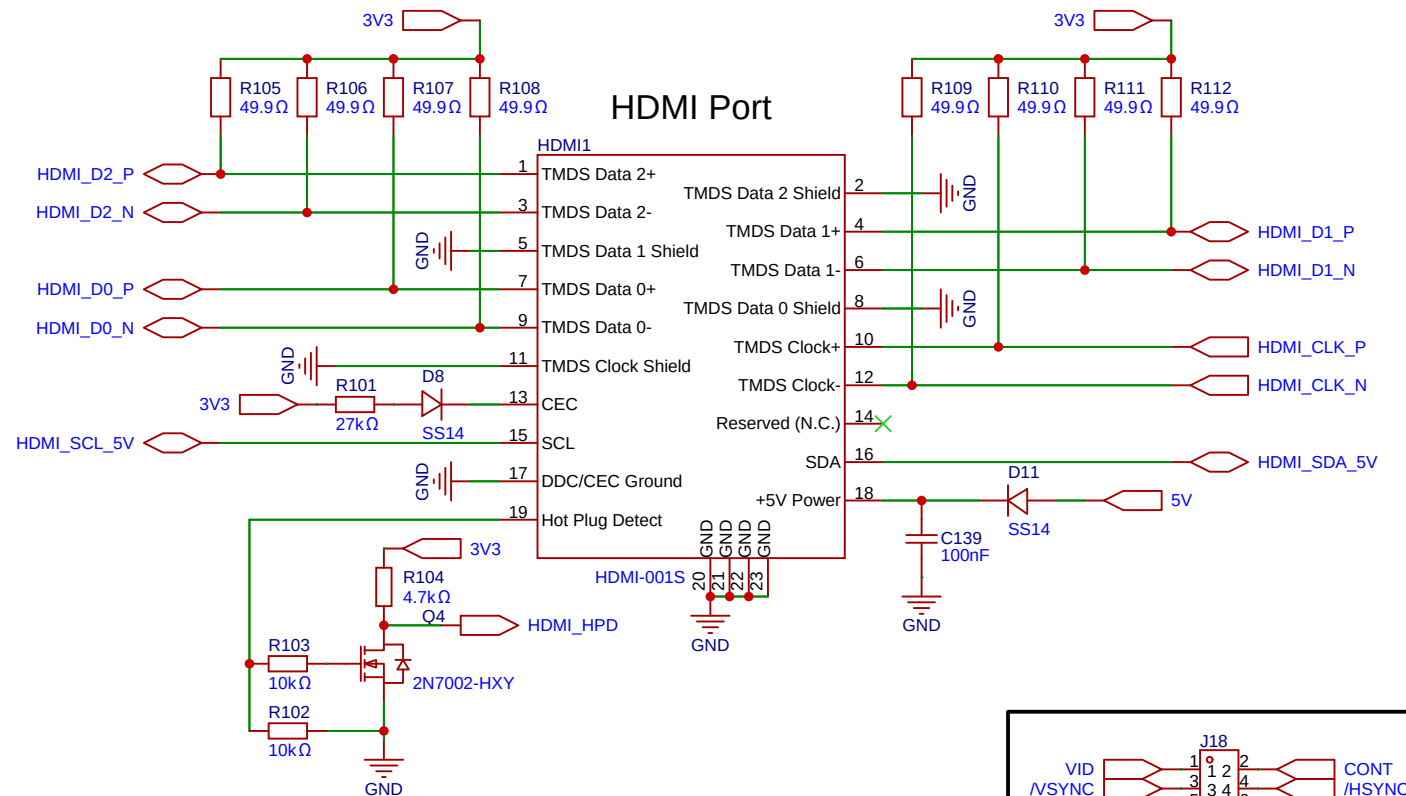


Physical Ports

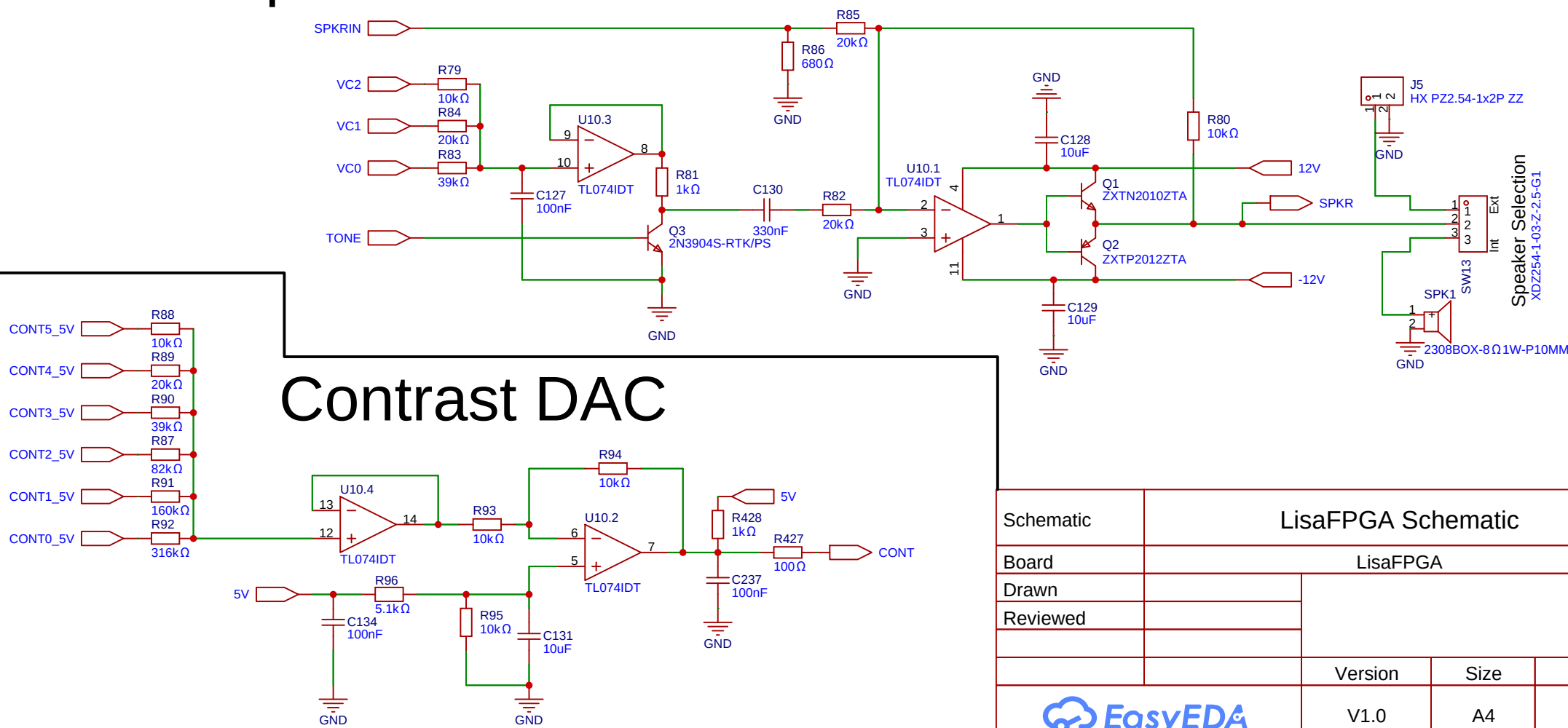


Schematic	LisaFPGA Schematic			Create at	2025-10-28
				Update at	2025-10-31
Board	LisaFPGA			Page	Serial Ports
Drawn	LisaFPGA				
Reviewed					
		Version	Size	Page 5 Total 8	
		V1.0	A4	EasyEDA.com	

I think the HDMI diff pairs need 100-ohm differential impedance when routing or something like that.

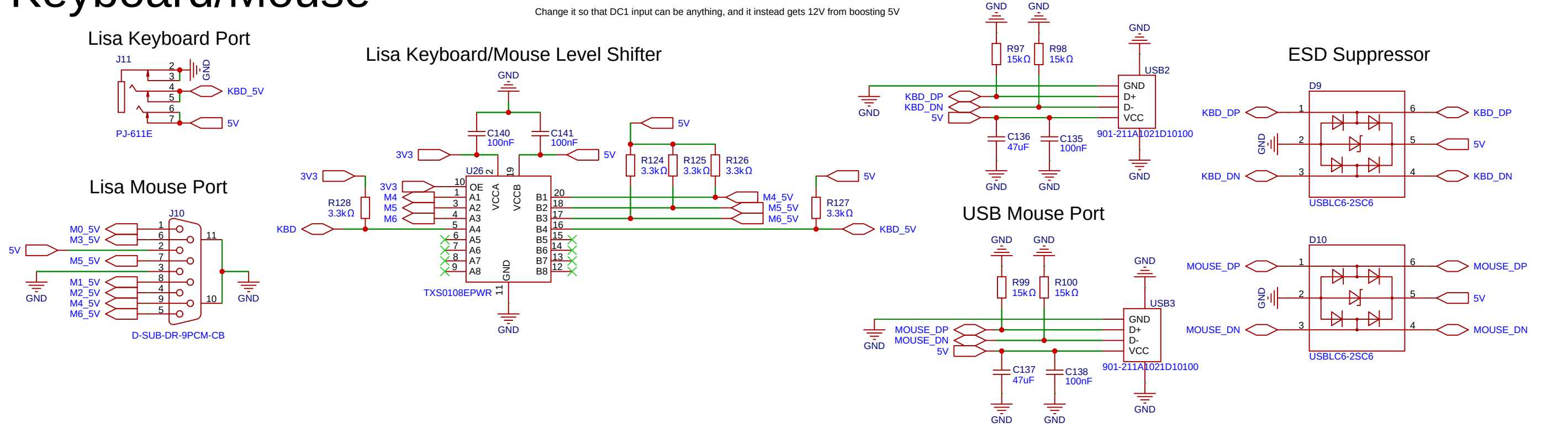


Contrast DAC

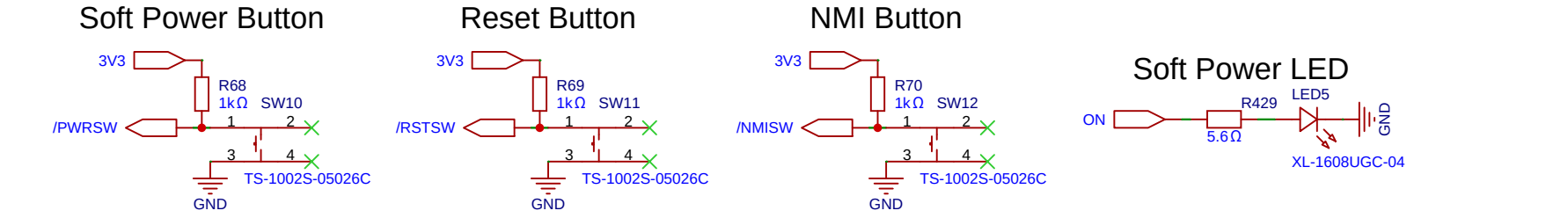


Schematic	LisaFPGA Schematic			Create at	2025-10-28
				Update at	2025-10-31
Board	LisaFPGA			Page	Audio/Video
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 6 Total 8	
		V1.0	A4	EasyEDA.com	

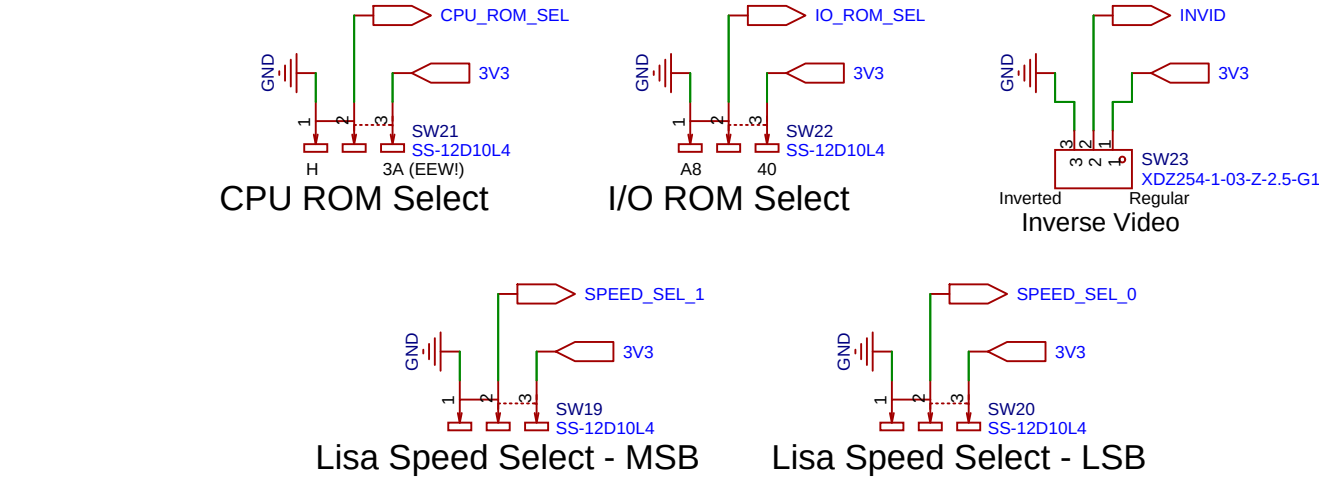
Keyboard/Mouse



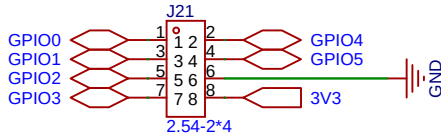
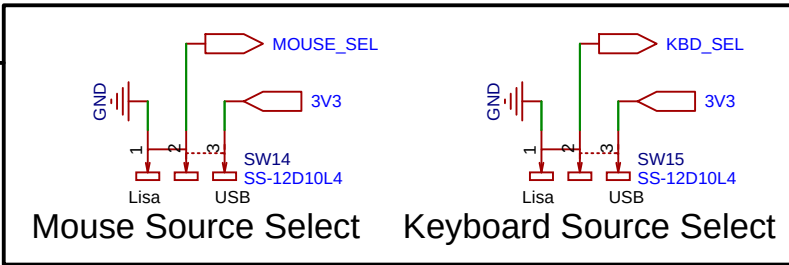
Lisa Buttons



Config Switches



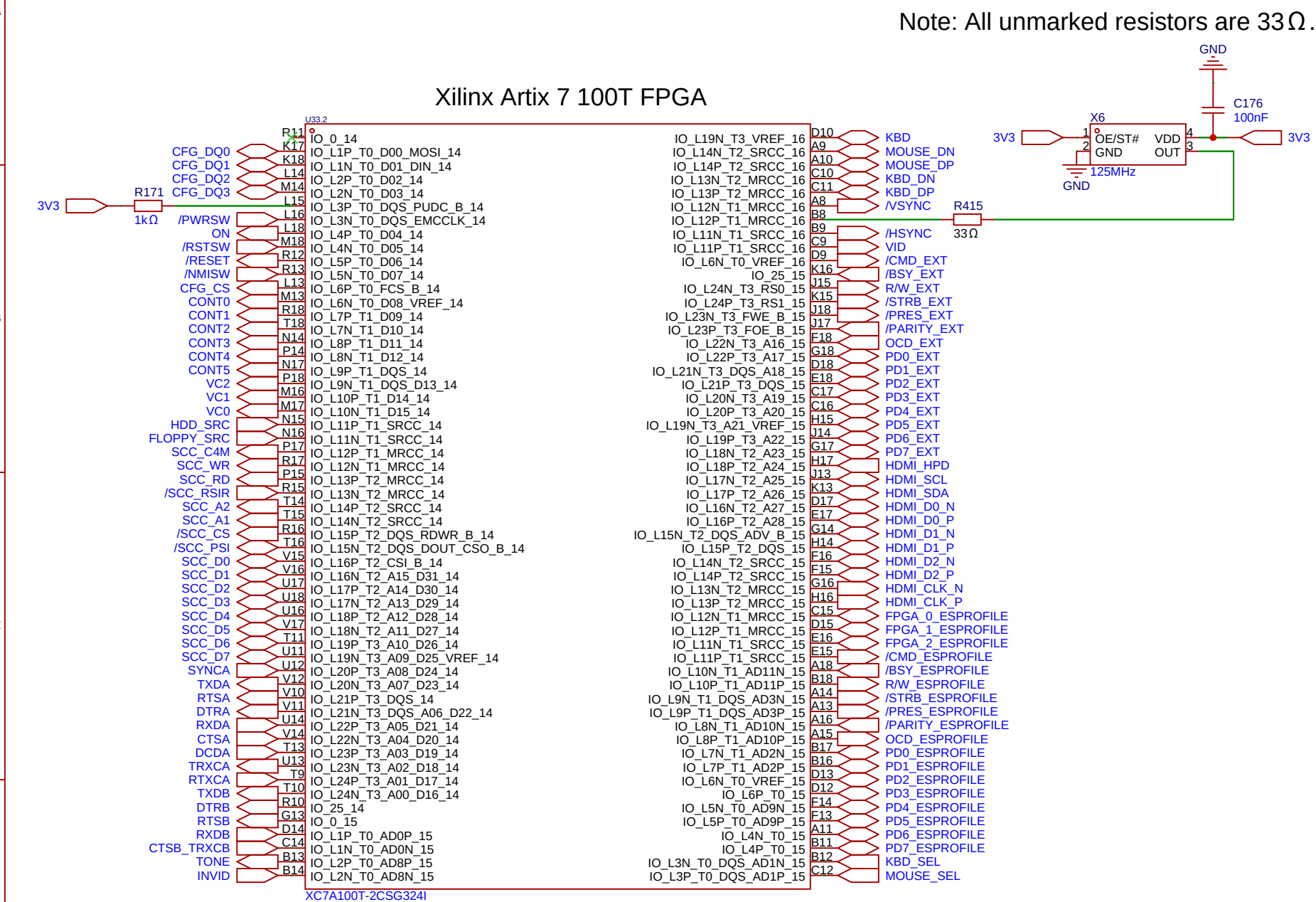
GPIO



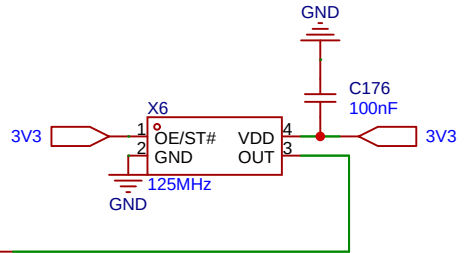
Wow, what a phenomenal use of space.
Looks like I screwed up my layout again...

Schematic	LisaFPGA Schematic			Create at	2025-10-28
				Update at	2025-11-01
Board	LisaFPGA			Page	Misc. I/O
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 7 Total 8	
EasyEDA			V1.0	A4	EasyEDA.com

FPGA I/O Connections



Note: All unmarked resistors are 33Ω .



Vias for bypass caps should be hooked straight to the side of the cap pads;
NO TRACE IN BETWEEN!!!
Going off of this, NEVER SHARE A VIA BETWEEN TWO CAPS!!!

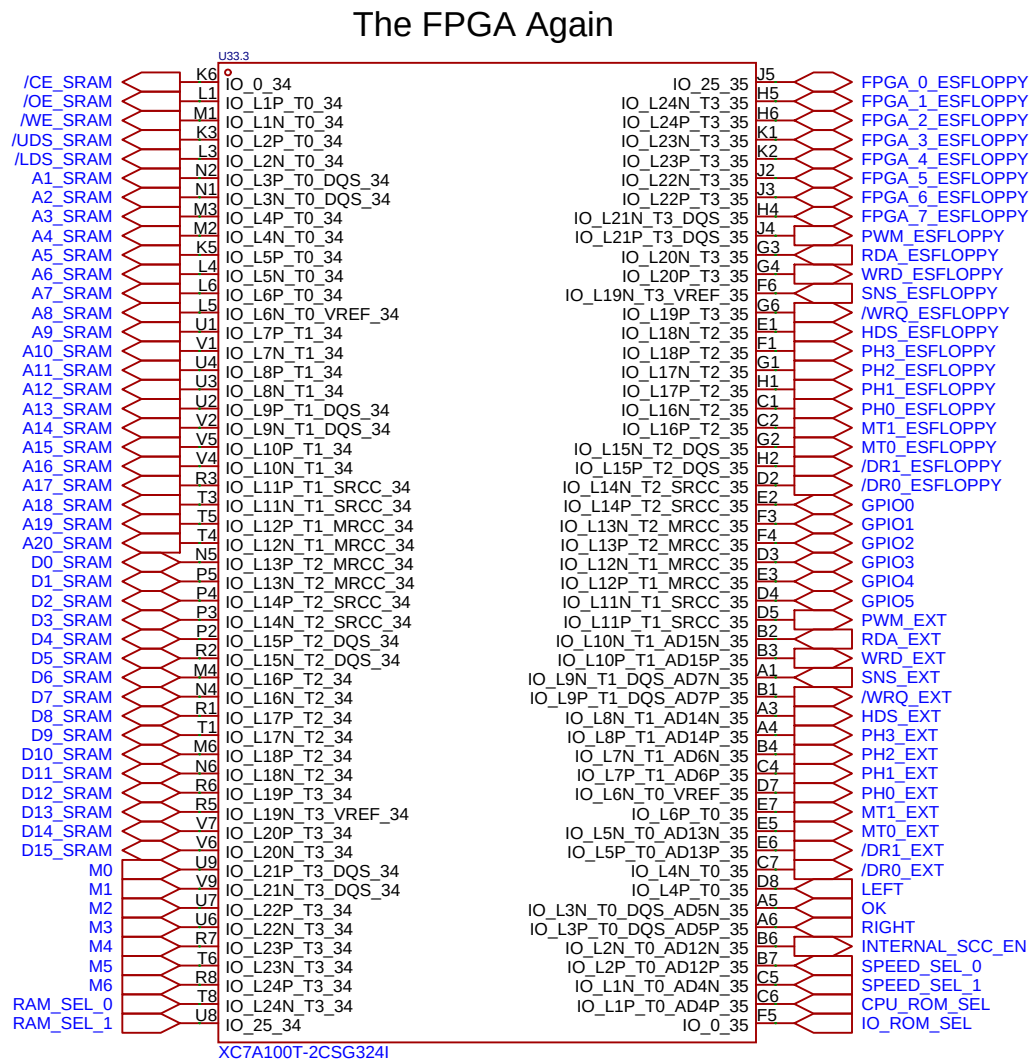
Term resistors should be closer to FPGA than all caps if you have to choose

Higher-priority power planes and those that carry more current should be higher in PCB stack

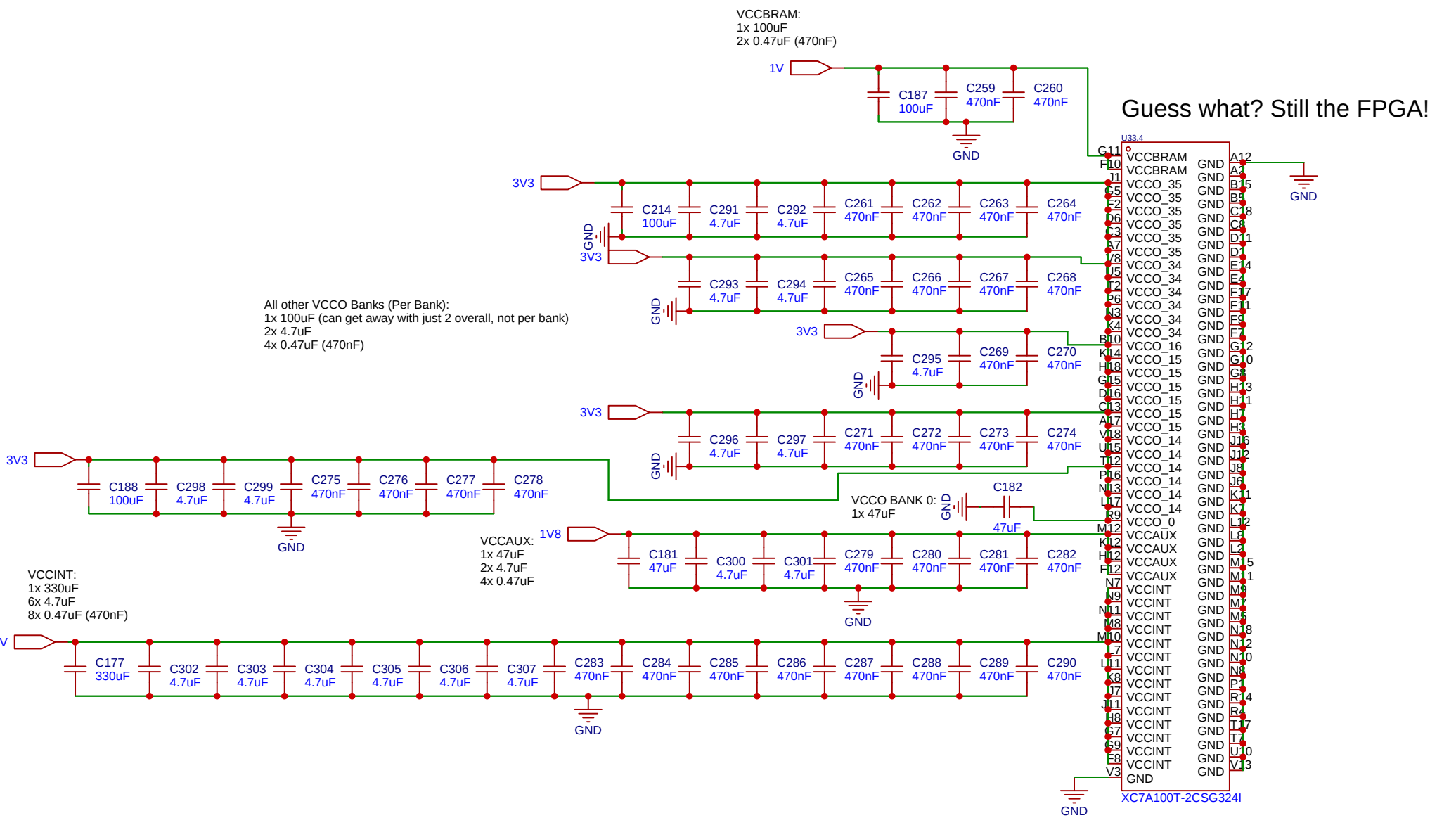
If the power plane is on the bottom half of the stack, put the caps on the bottom of the board

Put vias on the sides of cap pads, not the ends if possible

Bulk caps (all FPGA caps 100uF and larger) can go anywhere on PCB, but ideally put them close to FPGA; obviously all other ones must be close to FPGA



FPGA Power



Schematic	LisaFPGA Schematic			Create at	2025-10-28
				Update at	2025-11-03
Board	LisaFPGA			Page	FPGA Connections
Drawn		LisaFPGA			
Reviewed					
		Version	Size	Page 8 Total 8	
		V1.0	A4	EasyEDA.com	